INFRASTRUCTURE DELIVERY · PORTFOLIO CASE STUDY

Capacity Delivery Control Tower

An Executive Decision-Support Framework for Data Center Infrastructure Delivery

A six-page case study demonstrating how portfolio intelligence, schedule reconciliation, management accountability, and AI-assisted validation can support infrastructure delivery decisions.

15
Projects

544 MW
Approved capacity

5
Power BI report pages

EXECUTIVE CASE STUDY

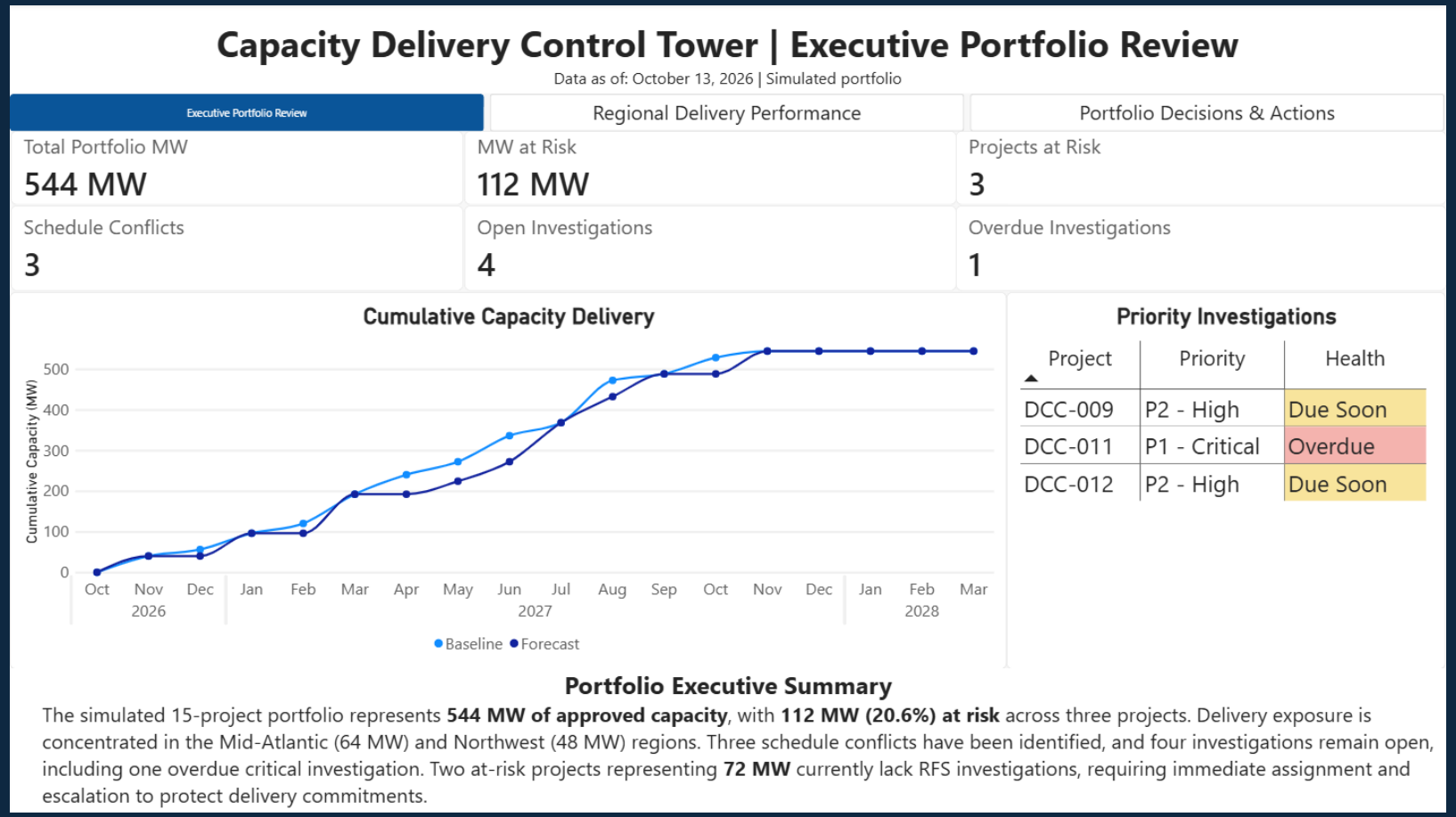
Capacity Delivery Control Tower

Executive Portfolio Intelligence for Data Center Infrastructure Delivery

544 MW
Approved capacity

112 MW
Capacity at risk

15
Projects



Key finding: 20.6% of approved capacity is at risk across three projects. The control tower links schedule exposure to priority investigations and executive decisions.

DELIVERY RISK & ACCOUNTABILITY

From Regional Exposure to Management Action

72 MW
At-risk capacity without RFS investigations

5
Executive actions tracked

Regional Delivery Performance

Where is capacity at risk, and who needs to act?

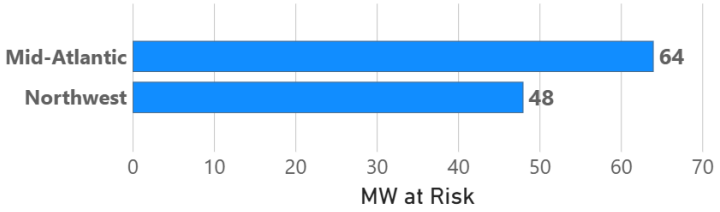
Data as of: October 13, 2026 | Simulated portfolio

Executive Portfolio Review | Regional Delivery Performance | Portfolio Decisions & Actions

Capacity at Risk	Projects at Risk	Portfolio Exposure	Missing RFS Investigations	Risk Without Investigation
112 MW	3 / 15	20.6%	2	72 MW

RFS Capacity at Risk by Region

(>21 Days)



Regional Intervention Summary

Projects Requiring Escalation

Region	Projects	MW
Northwest	1	48
Mid-Atlantic	1	24
Total	**2**	**72**

At-Risk Projects — Intervention List

Project	Region	MW	RFS Delay	Owner	RFS Investigation	Due Date	Escalation
DCC-007	Northwest	48	42	Avery Chen	No RFS Investigation		Escalate - No Investigation
DCC-004	Mid-Atlantic	24	35	Jordan Patel	No RFS Investigation		Escalate - No Investigation
DCC-012	Mid-Atlantic	40	28	Avery Chen	Investigation Recorded	10/15/2026	Due Soon

Portfolio Decisions & Actions

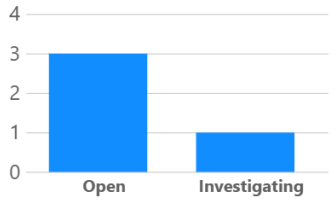
What requires action, who owns it, and when must it be resolved?

Data as of: October 13, 2026 | Simulated portfolio

Executive Portfolio Review | Regional Delivery Performance | Portfolio Decisions & Actions

Active Executive Actions	Actions Investigating	Overdue Investigations	Missing RFS Investigation
5	1	1	2

Investigation Action Pipeline



Action Ownership & Deadlines

Project	Milestone	Priority	Owner	Status	Due Date	Days Left	Health
DCC-011	FA	P1 - Critical	Regional Delivery Manager	Investigating	10/12/2026	-1	Overdue
DCC-009	RFS	P2 - High	GC Project Manager	Open	10/15/2026	2	Due Soon
DCC-012	RFS	P2 - High	GC Project Manager	Open	10/15/2026	2	Due Soon
DCC-004	FA	P3 - Review	Developer PM	Open	10/19/2026	6	On Track

Executive Action Register

Action	Project	Milestone	Required Action	Owner	Due Date	Status	Escalation
ACT-001	DCC-011	FA	Escalate overdue investigation	Regional Delivery Manager	10/12/2026	Investigating	Overdue
ACT-002	DCC-009	RFS	Confirm resolution plan before deadline	GC Project Manager	10/15/2026	Open	Due Soon
ACT-003	DCC-012	RFS	Confirm resolution plan before deadline	GC Project Manager	10/15/2026	Open	Due Soon
ACT-004	DCC-007	RFS	Initiate RFS investigation	Avery Chen		Open	Assign Due Date
ACT-005	DCC-004	RFS	Initiate RFS investigation	Jordan Patel		Open	Assign Due Date

Note: 72 MW is included within the 112 MW at risk

Management finding: DCC-007 and DCC-004 require RFS investigation initiation, while an overdue critical investigation requires escalation.

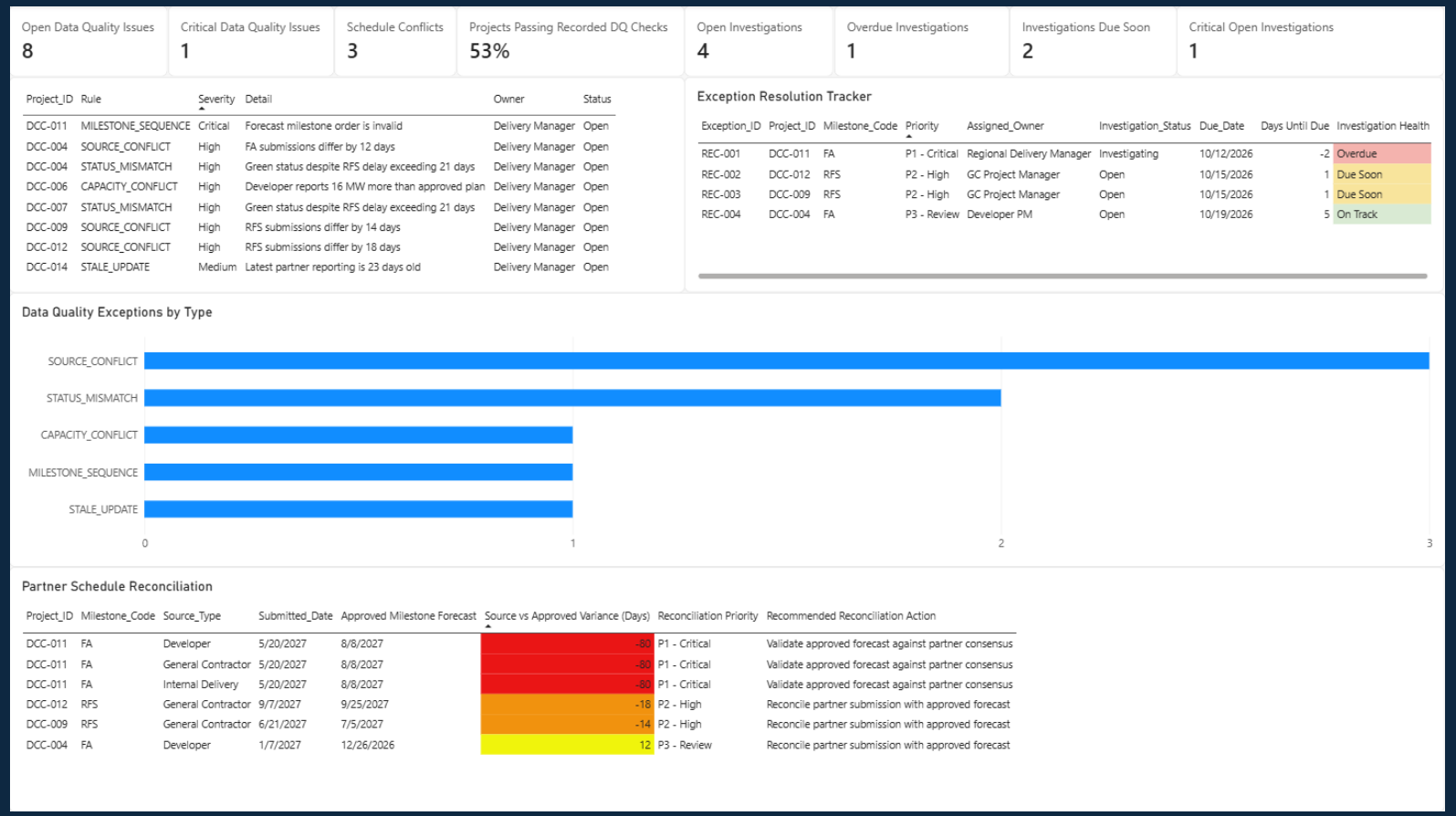
DATA QUALITY & RECONCILIATION

Establishing Confidence in Delivery Data

8
Open DQ Issues

3
Source Conflicts

1
Critical DQ Issue



Open Data Quality Issues	Critical Data Quality Issues	Schedule Conflicts	Projects Passing Recorded DQ Checks	Open Investigations	Overdue Investigations	Investigations Due Soon	Critical Open Investigations
8	1	3	53%	4	1	2	1

Project_ID	Rule	Severity	Detail	Owner	Status
DCC-011	MILESTONE_SEQUENCE	Critical	Forecast milestone order is invalid	Delivery Manager	Open
DCC-004	SOURCE_CONFLICT	High	FA submissions differ by 12 days	Delivery Manager	Open
DCC-004	STATUS_MISMATCH	High	Green status despite RFS delay exceeding 21 days	Delivery Manager	Open
DCC-006	CAPACITY_CONFLICT	High	Developer reports 16 MW more than approved plan	Delivery Manager	Open
DCC-007	STATUS_MISMATCH	High	Green status despite RFS delay exceeding 21 days	Delivery Manager	Open
DCC-009	SOURCE_CONFLICT	High	RFS submissions differ by 14 days	Delivery Manager	Open
DCC-012	SOURCE_CONFLICT	High	RFS submissions differ by 18 days	Delivery Manager	Open
DCC-014	STALE_UPDATE	Medium	Latest partner reporting is 23 days old	Delivery Manager	Open

Exception Resolution Tracker

Exception_ID	Project_ID	Milestone_Code	Priority	Assigned_Owner	Investigation_Status	Due_Date	Days Until Due	Investigation Health
REC-001	DCC-011	FA	P1 - Critical	Regional Delivery Manager	Investigating	10/12/2026	-2	Overdue
REC-002	DCC-012	RFS	P2 - High	GC Project Manager	Open	10/15/2026	1	Due Soon
REC-003	DCC-009	RFS	P2 - High	GC Project Manager	Open	10/15/2026	1	Due Soon
REC-004	DCC-004	FA	P3 - Review	Developer PM	Open	10/19/2026	5	On Track

Data Quality Exceptions by Type

Partner Schedule Reconciliation

Project_ID	Milestone_Code	Source_Type	Submitted_Date	Approved Milestone Forecast	Source vs Approved Variance (Days)	Reconciliation Priority	Recommended Reconciliation Action
DCC-011	FA	Developer	5/20/2027	8/8/2027	-80	P1 - Critical	Validate approved forecast against partner consensus
DCC-011	FA	General Contractor	5/20/2027	8/8/2027	-80	P1 - Critical	Validate approved forecast against partner consensus
DCC-011	FA	Internal Delivery	5/20/2027	8/8/2027	-80	P1 - Critical	Validate approved forecast against partner consensus
DCC-012	RFS	General Contractor	9/7/2027	9/25/2027	-18	P2 - High	Reconcile partner submission with approved forecast
DCC-009	RFS	General Contractor	6/21/2027	7/5/2027	-14	P2 - High	Reconcile partner submission with approved forecast
DCC-004	FA	Developer	1/7/2027	12/26/2026	12	P3 - Review	Reconcile partner submission with approved forecast

Note: 53% recorded DQ-check pass rate is separate from the 42-of-45 Power Query validation pass rate.

The control system distinguishes cross-source milestone disagreements from differences between submitted dates and approved portfolio forecasts. Exceptions are linked to investigation ownership, priorities, and deadlines.

AI VALIDATION & GOVERNANCE

Evidence-Backed AI With Human Oversight

45	3	0
Milestone Checks	Detected Conflicts	AI Briefs Approved

AI Briefs Generated	AI Briefs Awaiting Review	AI Briefs Needing Revision	AI Briefs Approved
3	2	1	0

Source Conflict & AI Review Traceability

Project	Milestone	Variance (Days)	Earliest Source	Latest Source	Investigation	Human Review Status
DCC-004	FA	12	SUB-0031 | Internal Delivery; SUB-0032 | General Contractor	SUB-0033 | Developer	DQ-001	Needs Revision
DCC-009	RFS	14	SUB-0080 | General Contractor	SUB-0079 | Internal Delivery; SUB-0081 | Developer	DQ-002	Awaiting Review
DCC-012	RFS	18	SUB-0107 | General Contractor	SUB-0106 | Internal Delivery; SUB-0108 | Developer	DQ-003	Awaiting Review

Validation & AI Governance Methodology

Source submissions are evaluated against a configurable 7-day milestone variance threshold using the latest eligible submission from each reporting source as of the selected reporting date. Conflicts are linked to Copilot-assisted investigation drafts and tracked through human review. AI-generated findings are not treated as authoritative until independently validated and approved.

Portfolio demonstration using simulated data. AI brief generation is prompt-triggered; validation refresh is initiated through Power Query.

Validated behavior: A simulated data change increased detected conflicts from three to four; restoring the source returned the count to three. A reporting-date cutoff also correctly excluded later-received submissions.

How the Data Is Managed and What Tools Were Used

From source submissions to executive insights, with validation, traceability, and human oversight.

Capacity Delivery Control Tower
Independent portfolio demonstration
Simulated data | October 13, 2026

1 Source Data Inputs
Multiple reporting sources

Internal Delivery
- Project updates
- Milestone dates
- Capacity changes
- Status and risks

General Contractors
- RFS/FA submissions
- Forecast updates
- Delivered capacity
- Comments and risks

Developers
- Milestone dates
- Capacity submissions
- Project status
- Supporting notes

Reference Data
- Approved portfolio forecast
- Project metadata
- Reporting calendar
- Validation rules

2 Data Preparation & Validation
Excel Tables and Power Query

Excel Tables
- Structured source files
- Consistent column mapping
- Historical submission storage

Power Query (M)
- Clean and standardize data
- Combine multiple sources
- Apply business rules
- Create traceability fields
- Filter to reporting date

Validation Engine
- 7-day milestone variance check
- Cross-source conflict detection
- Approved-forecast reconciliation
- Data-quality rule checks
- Flag exceptions and severity

3 Data Model & Calculations
Power BI (DAX)

Power BI Data Model
- Integrated, validated dataset
- DAX calculations for capacity, risk, and status
- Time-intelligent measures
- Investigation and action tracking
- Dynamic filtering by region, project, and milestone

Key Model Components
- **Projects** — Region, capacity, status
- **Milestones** — RFS, FA, COD dates
- **Submissions** — Source, date, version
- **Investigations** — Status, owner, due date
- **Executive Actions** — Escalation and ownership

4 Power BI Outputs
Executive decision-support dashboards



Executive Portfolio Review
Portfolio capacity, delivery forecast, and priority investigations



Regional Delivery Performance
Regional exposure and projects requiring escalation



Portfolio Decisions & Actions
Investigation pipeline, executive actions, owners, and deadlines



Data Quality & Reconciliation
Data-quality issues, source conflicts, and partner schedule comparison



AI Validation
Detected source conflicts and human review status

5 AI-Assisted Investigation
Microsoft Copilot with human oversight

Copilot Investigation Drafts
- Summarize source differences
- Identify potential impact
- Suggest investigation questions
- Recommend next steps

Human Review
- Validate source evidence
- Confirm authoritative dates
- Assess business impact
- Edit and approve findings

Approved Outcome
- Investigation updated
- Executive actions assigned
- Status tracked in Power BI

Point-in-Time Reporting
Configurable reporting date excludes later-received submissions to support reproducible historical analysis.

Data Traceability
All findings retain source submission IDs, source types, and version history.

Controlled Testing
Validated by introducing a simulated source change (3 → 4 conflicts) and testing the reporting-date cutoff.

Human Governance
AI-generated findings are not treated as authoritative until independently validated and approved by a human reviewer.

ARCHITECTURE & GOVERNANCE NOTE

The diagram illustrates the intended workflow, not a fully automated or completed approval process. As of October 13, 2026, three Copilot-assisted investigation briefs had been generated: two awaiting review, one needing revision, and zero approved. Power Query refreshes and Copilot drafting were manually initiated.